NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-83532/3A
30 December 1991
SUPERSEDING
MIL-D-83532/3
31 March 1987

PERFORMANCE SPECIFICATION SHEET

DELAY LINES, 14-PIN DIP COMPATIBLE, 10 TAP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-D-83532.

REQUIREMENTS:

Design and construction: Parts shall be of the design, construction, and physical dimensions specified in the appendix of MIL-D-83532. Four case styles (A, B, C, and D) are available.

Input pulse: Delay lines must be capable of meeting applicable table I and table II requirements with an input pulse having the following characteristics:

A: Leading edge of a positive-going pulse.

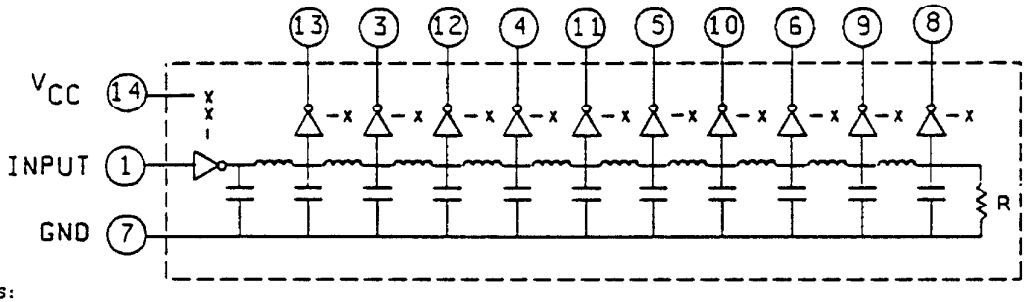
B: Minimum pulse width of 50 percent of total delay time.

C: Fixed pulse repetition rate equal to ten times the total delay time.

D: Duty cycle not to exceed 50 percent.

Schematic: See figure 1.

Delay times: Delay time from input to all taps shall be as specified in table I (+25°C) and table II (-55°C and +125°C).



NOTES:
1. Schematic is for general information only.
2. Pin 2 is omitted on level A delay lines.
3. Pin 2 is installed on level B delay lines.

FIGURE 1. Schematic.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Dash numbers anc delay characteristics at +25°C, V_{CC} = 5.00 ±.01 volts.

Dash number	Delay and tolerances (ns)									
	Tap 1 Pin 13	Tap 2 Pin 3	Tap 3 Pin 12	Tap 4 Pin 4	Tap 5 Pin 11	Tap 6 Pin 5	Tap 7 Pin 10	Tap 8 Pin 6	Tap 9 Pin 9	Tap 10 Pin 8
001	5 ±2 ns	10 ±2 rs	15 ±2 ns	20 ±2 ns	25 ±2 ns	30 ±2 ns	35 ±2 ns	40 ±2 ns	45 ±5%	50 ±5%
002	7.5 ±2 ns	15 ±2 rs	22.5 ±2 ns	30 ±2 ns	37.5 ±2 ns	45 ±5%	52.5 ±5%	60 ±5%	67.5 ±5%	75 ±5%
003	10 ±2 ns	20 ±2 rs	30 ±2 ns	40 ±2 ns	50 ±5%	60 ±5%	70 ±5%	80 ±5%	90 ±5%	100 ±5%
004	12.5 ±2 ns	25 ±2 rs	37.5 ±2 ns	50 ±5%	62.5 ±5%	75 ±5%	87.5 ±5%	100 ±5%	112.5 ±5%	125 ±5%
005	15 ±2 ns	30 ±2 rs	45 ±5%	60 ±5%	75 ±5%	90 ±5%	105 ±5%	120 ±5%	135 ±5%	150 ±5%
006	17.5 ±2 ns	35 ±2 rs	52.5 ±5%	70 ±5%	87.5 ±5%	105 ±5%	122.5 ±5%	140 ±5%	157.5 ±5%	175 ±5%
007	20 ±2 ns	40 ±2 rs	60 ±5%	80 ±5%	100 ±5%	120 ±5%	140 ±5%	160 ±5%	180 ±5%	200 ±5%
008	25 ±2 ns	50 ±5%	75 ±5%	100 ±5%	125 ±5%	150 ±5%	175 ±5%	200 ±5%	225 ±5%	250 ±5%
009	30 ±2 ns	60 ±5%	90 ±5%	120 ±5%	150 ±5%	180 ±5%	210 ±5%	240 ±5%	270 ±5%	300 ±5%
010	35 ±2 ns	70 ±5%	105 ±5%	140 ±5%	175 ±5%	210 ±5%	245 ±5%	280 ±5%	315 ±5%	350 ±5%
011	40 ±2 ns	80 ±5%	120 ±5%	160 ±5%	200 ±5%	240 ±5%	280 ±5%	320 ±5%	360 ±5%	400 ±5%
012	45 ±5%	90 ±5%	135 ±5%	180 ±5%	225 ±5%	270 ±5%	315 ±5%	360 ±5%	405 ±5%	450 ±5%
013	50 ±5%	100 ±5%	150 ±5%	200 ±5%	250 ±5%	300 ±5%	350 ±5%	400 ±5%	450 ±5%	500 ±5%
014	60 ±5%	120 ±5%	180 ±5%	240 ±5%	300 ±5%	360 ±5%	420 ±5%	480 ±5%	540 ±5%	600 ±5%
015	70 ±5%	140 ±5%	210 ±5%	280 ±5%	350 ±5%	420 ±5%	490 ±5%	560 ±5%	630 ±5%	700 ±5%
016	80 ±5%	160 ±5%	240 ±5%	320 ±5%	400 ±5%	480 ±5%	560 ±5%	640 ±5%	720 ±5%	800 ±5%
017	90 ±5%	180 ±5%	270 ±5%	360 ±5%	450 ±5%	540 ±5%	630 ±5%	720 ±5%	810 ±5%	900 ±5%
018	100 ±5%	200 ±5%	300 ±5%	400 ±5%	500 ±5%	600 ±5%	700 ±5%	800 ±5%	900 ±5%	1000 ±5%

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TABLE II. Dash numbers and delay characteristics at -55°C and +125°C, V_{CC} = 5.00 ±.01 volts.

Dash number	Delay and tolerances (ns)									
	Tap 1 Pin 13	Tap 2 Pin 3	Tap 3 Pin 12	Tap 4 Pin 4	Tap 5 Pin 11	Tap 6 Pin 5	Tap 7 Pin 10	Tap 8 Pin 6	Tap 9 Pin 9	Tap 10 Pin 8
001	5 ±3 ns	10 ±3 ns	15 ±3 ns	20 ±3 ns	25 ±3 ns	30 ±3 ns	35 ±3 ns	40 ±3 ns	45 ±8%	50 ±8%
002	7.5 ±3 ns	15 ±3 ns	22.5 ±3 ns	30 ±3 ns	37.5 ±3 ns	45 ±8%	52.5 ±8%	60 ±8%	67.5 ±8%	75 ±8%
003	10 ±3 ns	20 ±3 ns	30 ±3 ns	40 ±3 ns	50 ±8%	60 ±8%	70 ±8%	80 ±8%	90 ±8%	100 ±8%
004	12.5 ±3 ns	25 ±3 ns	37.5 ±3 ns	50 ±8%	62.5 ±8%	75 ±8%	87.5 ±8%	100 ±8%	112.5 ±8%	125 ±8%
005	15 ±3 ns	30 ±3 ns	45 ±8%	60 ±8%	75 ±8%	90 ±8%	105 ±8%	120 ±8%	135 ±8%	150 ±8%
006	17.5 ±3 ns	35 ±3 ns	52.5 ±8%	70 ±8%	87.5 ±8%	105 ±8%	122.5 ±8%	140 ±8%	157.5 ±8%	175 ±8%
007	20 ±3 ns	40 ±3 ns	60 ±8%	80 ±8%	100 ±8%	120 ±8%	140 ±8%	160 ±8%	180 ±8%	200 ±8%
008	25 ±3 ns	50 ±8%	75 ±8%	100 ±8%	125 ±8%	150 ±8%	175 ±8%	200 ±8%	225 ±8%	250 ±8%
009	30 ±3 ns	60 ±8%	90 ±8%	120 ±8%	150 ±8%	180 ±8%	210 ±8%	240 ±8%	270 ±8%	300 ±8%
010	35 ±3 ns	70 ±8%	105 ±8%	140 ±8%	175 ±8%	210 ±8%	245 ±8%	280 ±8%	315 ±8%	350 ±8%
011	40 ±3 ns	80 ±8%	120 ±8%	160 ±8%	200 ±8%	240 ±8%	280 ±8%	320 ±8%	360 ±8%	400 ±8%
012	45 ±8%	90 ±8%	135 ±8%	180 ±8%	225 ±8%	270 ±8%	315 ±8%	360 ±8%	405 ±8%	450 ±8%
013	50 ±8%	100 ±8%	150 ±8%	200 ±8%	250 ±8%	300 ±8%	350 ±8%	400 ±8%	450 ±8%	500 ±8%
014	60 ±8%	120 ±8%	180 ±8%	240 ±8%	300 ±8%	360 ±8%	420 ±8%	480 ±8%	540 ±8%	600 ±8%
015	70 ±8%	140 ±8%	210 ±8%	280 ±8%	350 ±8%	420 ±8%	490 ±8%	560 ±8%	630 ±8%	700 ±8%
016	80 ±8%	160 ±8%	240 ±8%	320 ±8%	400 ±8%	480 ±8%	560 ±8%	640 ±8%	720 ±8%	800 ±8%
017	90 ±8%	180 ±8%	270 ±8%	360 ±8%	450 ±8%	540 ±8%	630 ±8%	720 ±8%	810 ±8%	900 ±8%
018	100 ±8%	200 ±8%	300 ±8%	400 ±8%	500 ±8%	600 ±8%	700 ±8%	800 ±8%	900 ±8%	1000 ±8%

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TABLE III. DC characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Limits		Unit
			Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V V_{IH} = 2.0 V I_{OH} = -1 mA	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V V_{IL} = 0.8 V I_{OL} = 20 mA		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V I_I = -18 mA T_C = +25°C		-1.2	V
High level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IH} = 2.7 V		100	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IH} = 5.5 V		2000	µA
Low level input current	I_{IL}	V_{CC} = 5.5 V, V_{IL} = 0.5		-4.0	mA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, V_{OS} = 0.0 V (not more than one output shorted at a time)	-40	-150	mA
Low level supply current	I_{CCL}	V_{CC} = 5.5 V V_I = 0.0 V		130	mA

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Output rise time (applied to leading edge only): 4 ns maximum for dash numbers 001 through 009; 5 ns maximum for dash numbers 010 through 015; and 6 ns maximum for 016 through 018. Measurement conditions ($-55°C \leq T_C \leq +125°C$): V_{CC} = 5.0 V dc; $TR_I \leq 3$ ns; C_L = 50 pF; R_L = 500Ω.

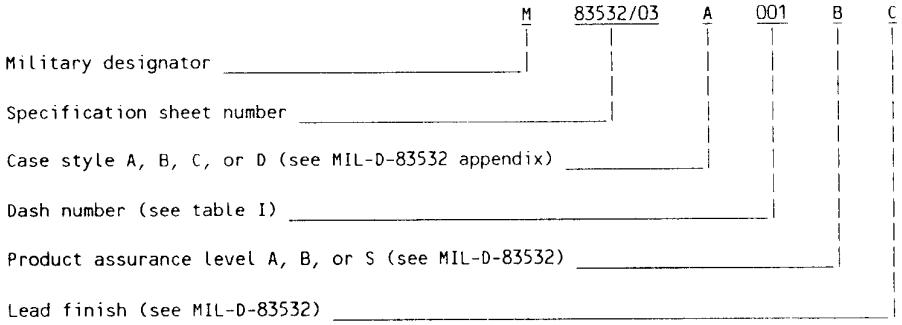
DC parameters (over operating temperature range): See table III.

Rated maximum load (fan-out): Ten TTL Schottky loads per tap (no more than 20 TTL Schottky loads per unit).

Terminations: Part terminations shall be printed-circuit pin type in accordance with MIL-H-38534.

Operating temperature: Operating temperature range shall be -55°C to +125°C.

Part or Identifying Number (PIN): The PIN shall be in the following format.



QUALITY ASSURANCE PROVISIONS:

Extent of qualification: The extent of qualification shall be as specified in MIL-D-83532, for example:

Qualification and testing of M83532/03A001A* 1/ and M83532/03A018A* 1/ shall be sufficient to grant qualification to all dash numbers with case style A contained in this specification sheet. Similarly, for case styles B, C, and D, qualification and testing of dash numbers 001 and 018 shall be sufficient to grant qualification to all dash numbers in this specification sheet having the same case style as the components tested.

Product assurance level: Product assurance level A shall not extend to level B. Level B shall not extend to level A.

Supersession data: MIL-D-83532/3 supersedes DESC drawing number 84014 when a qualified source is available.

Revision letters are not used to denote changes due to the extensiveness of the changes.

1/ * - Asterisk represents any of the applicable finishes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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